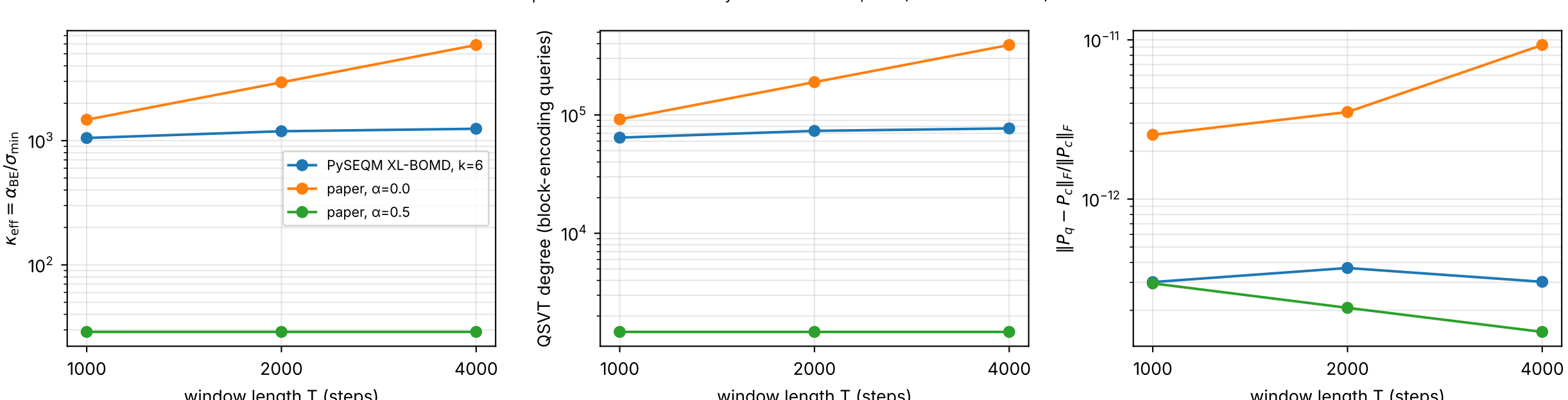
QSVT linear solve of the linearized XL-BOMD system, γ = 0.5
input: SYNTHETIC random symmetric matrices, d=4 (not molecular data)
$\kappa_{\mathrm{eff}} = \alpha_{\mathrm{BE}}/\sigma_{\min}$
10^3
10^2
PySEQM XL-BOMD, k=6
paper, α=0.0
paper, α=0.5
1000
2000
4000
window length T (steps)
QSVT degree (block-encoding queries)
10^5
10^4
1000
2000
4000
window length T (steps)
$\|P_q - P_c\|_F / \|P_c\|_F$
10^{-11}
10^{-12}
1000
2000
4000
window length T (steps)